# 2022 SSMO Accuracy Round

SMO Team

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**AR 1:** A bijective function $f : S \to S$, where $S = \{1, 2, 3, 4, 5\}$, is chosen uniformly at random. Find the expected value of $f(1) + f(2) + f(3)$.

**AR 2:** Consider a cone with radius 5 and height 12, and a point $P$ in the same plane as the base of the cone, but a distance of 10 from the center of the base of the cone. We rotate the cone 360° about $P$ such that the plane that the base of the cone lies on stays the same. The volume of the region that the cone sweeps out can be expressed as $m\pi$. Find $m$.

**AR 3:** Let $A = (6, 3, 2)$, $B = (2, -9, -6)$, and $O = (0, 0, 0)$. Suppose that $D$ is a point in space such that $OD$ bisects $\angle AOB$ and $O, D, A, B$ are coplanar. In addition, $\angle DAO = 90°$. The length of $DO$ can be expressed as $\frac{a\sqrt{b}}{c}$, where $a$, $b$, and $c$ are positive integers, $a$ and $c$ are relatively prime, and $b$ is not divisible by the square of any prime. Find $a + b + c$.

**AR 4:** A monic polynomial $f$ has real roots $r, s, t$ and a monic polynomial $g$ has roots $r^3, s^3, t^3$. The minimum value of $\frac{g(1)}{f(1)}$ can be expressed as $\frac{m}{n}$, where $m$ and $n$ are relatively prime positive integers. Find $m + n$.

**AR 5:** Find the number of ordered pairs $(a, b)$, where $1 \leq a, b \leq 9$, for which the largest integer $n$ that satisfies

$$(a - b)^k \equiv a^k - b^k \pmod{n}$$

for all $k \geq 1$ is $ab - b^2$.

**AR 6:** Consider an unfair 6-sided die labeled from 1 to 6, such that the probability of rolling a number $m$ is directly proportional to $7 - m$. However, if we roll any number $n$, then the probability of rolling a number less than $n$ becomes 0, and the probability of rolling any number $m$ from $n$ to 6 inclusive remains directly proportional to $7 - m$. The expected number of rolls until a 6 is rolled can be expressed as $\frac{m}{n}$, where $m$ and $n$ are relatively prime positive integers. Find $m + n$.

**AR 7:** After a robber drives in a car for $t$ (not necessarily integral) minutes, the car goes at $120 - t$ miles per hour. Whenever the car's speed drops below 60 miles per hour, the robber switches into a new car with no time loss. A police car can drive at a constant speed of 117 miles per hour. Given that the robber starts 1 hour before the police car, how many minutes will pass between when the police car starts and when the police car catches up to the robber?

**AR 8:** Let $ABCD$ be a trapezoid with $AB \parallel CD$. Suppose that $AD = 1$, $DC = 4$, $CB = 2$, and $AB < CD$. Let $X$ be the midpoint of $AB$ and $E$ denote the intersection of $AC$ and $BD$. If $\angle XEB = \angle ADC$, then $AB$ can be expressed as $\frac{m}{n}$, where $m$ and $n$ are relatively prime positive integers. Find $m + n$.

**AR 9:** The graph $\Gamma$ has 52 vertices labeled $A, A', B, B', \ldots, Z, Z'$, such that $A$ is not connected to $A'$, $B$ is not connected to $B'$, and so on. Suppose that all the vertices other than $A'$ have different degrees (number of connections to the vertex). Find the sum of all possible values for the number of edges (connections) in $\Gamma$.

**AR 10:** Let

$$S = \sum_{k=5}^{\infty} 2^{-10k} \binom{2k}{10}.$$

Then the value of $S$ can be expressed as $\frac{m}{n}$, where $m$ and $n$ are relatively prime positive integers. Find the largest positive integer $a$ such that $2^a \mid mn$.